

undoverse.in

Architecture & Implementation Strategy

Prepared for: **72BPM, Trivandrum, Kerala**

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ADR-001: Platform Architecture for undoverse.in

undoverse.in is the central aggregator, launchpad, and community hub for the "undo" ecosystem of web projects (currentundo, kuzhiundo, damundo, and future entries). The platform simultaneously serves as:

- A **multi-tenant hosting layer** — running sub-projects under one domain
- A **creator economy platform** — profiles, tipping, analytics
- A **community governance system** — upvoting, submissions, open-source hub

The core tension: ship an MVP fast (Phase 1 in weeks) while choosing an architecture that doesn't require a complete rewrite when Phase 2 and 3 arrive.

Section 1: Core Architecture Decision

Decision: Monorepo + Next.js App Router + Vercel Edge

Rather than splitting into separate services prematurely, undoverse.in starts as a well-structured monorepo with clear module boundaries. This enables fast MVP delivery while allowing easy service extraction later.

Monorepo Structure

```
undoverse/ | apps/web (Next.js) | apps/docs | packages/ui | packages/db |  
packages/auth | turbo.json
```

Why Turborepo + monorepo?

- Sub-projects (kuzhiundo, damundo etc.) can live as packages in the same repo — easy to link, share components
- One CI/CD pipeline, one contributor entry point — critical for open-source community adoption
- Vercel deploys each app independently — separate deployments, one repo

Section 2: Technical Stack Recommendations

Layer	Choice	Reasoning
Frontend	Next.js 14+ (App Router) + Tailwind + shadcn/ui	SSR/SSG per route, streaming, server components — critical for SEO. shadcn/ui for accessible
Auth	Auth.js v5 (GitHub OAuth)	GitHub OAuth out of the box. Extensible to UPI/wallet identities in Phase 3.
Database	PostgreSQL via Neon (serverless) + Prisma	Serverless Postgres — no cold-start issues on Vercel Edge. DB branching for dev environments
API Layer	Next.js Route Handlers (+ tRPC optional)	Start lean with Route Handlers. Add tRPC for end-to-end type safety as API surface grows.

Hosting	Vercel Pro + Cloudflare DNS	Wildcard subdomain (*.undoverse.in) is first-class on Vercel Pro. Cloudflare for DDoS protection
Background Jobs	Inngest	Serverless-native event queue for analytics aggregation, upvote tallying, email notifications.
Search	Algolia (free tier) or Meilisearch	Start with Algolia free tier (10K records). Migrate to self-hosted Meilisearch on Railway when needed.
Storage	Cloudflare R2	Project screenshots, builder avatars. S3-compatible API, zero egress fees.
Analytics	PostHog (open-source)	Self-hostable. Product analytics + session recordings. Creator analytics dashboard built on PostHog.

Section 3: Multi-Tenant Subdomain Routing

The most technically complex part of Phase 1. Goal: **kuzhiundo.undoverse.in** routes to the kuzhiundo app while sharing the same Next.js codebase.

Approach: Next.js Middleware + Vercel Wildcard Domains

```
// middleware.ts (runs on Vercel Edge) export function middleware(req: NextRequest) {
const host = req.headers.get('host') ?? ''
const subdomain = host.split('.undoverse.in')[0]
if (subdomain && SUBPROJECT_MAP[subdomain]) {
return NextResponse.rewrite( // URL stays for the user
new URL(SUBPROJECT_MAP[subdomain] + req.nextUrl.pathname, req.url) )
}
return NextResponse.next() }
```

DNS setup: Add a wildcard CNAME **.undoverse.in* → *cname.vercel-dns.com* in Cloudflare. Vercel handles TLS termination automatically. When a new project is approved, add a row to the subprojects table — the middleware reads from Edge-compatible KV, so routing updates without a redeploy.

Section 4: Core Database Schema

id (UUID)
github_id
username
display_name
avatar_url
bio
upi_id
wallet_addr
created_at
id (UUID)
slug

name
tagline
description
repo_url
status
subdomain
created_at

project_id (FK)
user_id (FK)
role (builder/maintainer/contributor)

id (UUID)
project_id (FK)
user_id (FK)
created_at
UNIQUE(project_id, user_id)

id (UUID)
submitter_id (FK)
project_name
concept_url
repo_url
status
created_at

Section 5: Phased Implementation Plan

Phase 1 — MVP (Target: 4 Weeks)

Goal: Live, publicly accessible undoverse.in with the three existing projects listed, core visual identity, and subdomain routing working.

Week 1 — Foundation

- Initialize monorepo: `pnpm create turbo@latest`, add `apps/web`
- Configure Tailwind + shadcn/ui + base design tokens (brand colors, typography)
- Set up Neon PostgreSQL, run initial migration
- Deploy skeleton to Vercel, configure undoverse.in domain + wildcard `*.undoverse.in`
- Set up public GitHub repo, add CONTRIBUTING.md — open-source from day one

→ Deliverable: Live domain, 'coming soon' page with brand identity

Week 2 — Core Landing Page

- Build Unified Dashboard: project cards for `currentundo`, `kuzhiundo`, `damundo`
- Implement subdomain middleware routing
- Create project detail pages `/projects/[slug]`
- Add global search (Algolia free tier) indexing project names and taglines
- Seed database with the three initial projects

→ Deliverable: Full landing page live, all three projects accessible via subdomains

Week 3 — Live Status & Polish

- Build changelog/status feed (scrape GitHub releases via API, cache in DB)
- Add uptime badges (UptimeRobot free API)
- Mobile-responsive pass on all pages
- SEO: Open Graph images per project (Vercel/og), sitemap, robots.txt
- Set up PostHog analytics

→ Deliverable: Production-ready MVP, shareable on social

Week 4 — Buffer, QA & Launch

- Load testing, error boundary setup
- Write README.md, developer setup guide, architecture docs
- Soft launch to Kerala dev community (Kerala Developers Slack, local meetups)
- Gather feedback, triage issues

Phase 2 — Creator & Automation Layer (6–8 Weeks post-MVP)

Goal: GitHub OAuth login, builder profiles, project submission pipeline, automated subdomain routing.

Sprint 1 — Auth + Profiles (2 weeks)

- Integrate Auth.js with GitHub OAuth
- Build builder profile pages /builders/[username]
- Profile editor: bio, UPI ID, wallet address, social links
- Connect builders to projects via project_builders table

Sprint 2 — Project Submission Portal (2 weeks)

- Build /submit page — form for new 'undo' project proposals
- Admin review queue (simple /admin route, protected by role check)
- On approval: auto-provision subdomain entry in Vercel KV, no redeploy needed
- Email notification on submission status (Resend.com)

Sprint 3 — Ecosystem Analytics (2 weeks)

- Per-project analytics dashboard (pageviews, unique visitors from PostHog API)
- Upvote counts displayed on project cards, sortable
- GitHub integration: pull 'good first issue' issues from project repos
- Display contributor count, stars, last commit date on project cards

Phase 3 — Economy & Scaling (3 months post-Phase 2)

Micro-Tipping & Monetization

- UPI deep-link on builder profiles (QR code — zero integration, works immediately)
- Buy Me a Coffee / Ko-fi embed on builder profiles
- Web3: WalletConnect + wagmi for ETH/MATIC tipping (lower priority than UPI for Kerala audience)
- Tip history table for transparency

Community Layer

- Comment threads on projects (Giscus — GitHub Discussions-backed, free, open-source aligned)
- Reddit-style upvote feed sorted by 'hot' (Wilson score ranking)
- Hackathon/build sprint pages — time-boxed challenge listings
- Weekly 'Undo Digest' email newsletter (Resend + custom template)

Scaling Infrastructure

- Move background jobs to Inngest (analytics rollups, digest emails)
- Add Redis (Upstash serverless) for rate limiting on upvotes and submissions
- Migrate search to self-hosted Meilisearch on Railway if Algolia free tier exceeded

- Extract 'project submission + routing' as open-source package @undoverse/host

Section 6: Open-Source Strategy

Keeping undoverse.in open-source is the **strategic multiplier** for the entire platform.

- **Kerala dev community builds the platform with you** — every contributor becomes an evangelist
- **Projects submitted to undoverse.in can also contribute back** — two-way flywheel
- **Credibility with the global developer-creator audience** you're targeting

Recommended actions from day one:

- Public GitHub repo under the undoverse organization, MIT license for core platform
- CONTRIBUTING.md with clear 'Good First Issues' from week 1
- Ship a create-undo CLI scaffolder: `npx create-undo my-project`
- Individual 'undo' projects can choose their own license

Section 7: Cost Estimate (MVP → Phase 2)

Service	Plan	Monthly Cost
Vercel	Pro (required for wildcard domains)	\$20
Neon PostgreSQL	Free tier (3GB) → Scale as needed	\$0–\$19
Cloudflare	Free (DNS, R2 10GB)	\$0
Algolia	Free (10K records)	\$0
PostHog	Free (1M events/month)	\$0
Resend	Free (3K emails/month)	\$0
Upstash Redis	Free (10K commands/day)	\$0
TOTAL MVP		~\$20/month

Note: Vercel Pro is non-negotiable for wildcard subdomain support. Everything else runs on free tiers through Phase 1 and most of Phase 2.

Section 8: Key Risks & Mitigations

Risk	Likelihood	Impact	Mitigation
Subdomain routing complexity	Medium	High	Ship MVP without subdomains first (use /projects/[slug]) — add subdomains in Week 3
Low project submission quality	Medium	Medium	Manual curation in Phase 1–2; use upvoting as quality signal in Phase 3
Vercel lock-in	Low	Medium	All infra choices (Neon, R2, Inngest) are portable. Can migrate to Railway/Fly.io

Open-source security issues	Low	High	Require PRs to pass CI, add CodeQL scanning from day one, role-check admin routes
Kerala community adoption	Medium	High	Start with offline events (Trivandrum meetups), build in public — Kerala founder story is

Section 9: Immediate Next Steps (This Week)

- 1 Register undoverse.in on Namecheap/GoDaddy if not already done; point DNS to Cloudflare
- 2 Create GitHub org undoverse-in, initialize the monorepo with pnpm create turbo
- 3 Design the brand — choose a font pair and color palette; 'undo/reset' concept has great arrow/reset iconography potential
- 4 Collect proper info for the first 3 projects — descriptions, repo URLs, builder names for currentundo, kuzhiundo, damundo
- 5 Upgrade Vercel to Pro — needed for wildcard domains; start 14-day trial now so it's ready when routing is built

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